HHL Quantum Linear-Systems Algorithm

Assessment of the Harrow–Hassidim–Lloyd algorithm

QUESTION PAPER

Time allowed:	3 hours	**Maximum marks:**	240
Questions:	80 (4 sections × 20)	**Answer key:**	separate Answer Book

Name of the candidate: _____ **Roll No.:** _____

General instructions

1. This question paper has **four sections** of **20 questions** each: **A** Linear Algebra and Mathematical Foundations, **B** Quantum Computing Foundations, **C** HHL Algorithm Fundamentals, **D** Advanced HHL and Implementation.
2. Every question is of the **single-correct** type: four options, exactly one correct. Question formats used: single-correct MCQ, matching lists (List-I/List-II), Assertion–Reason, Statement I/II and paragraph-based questions.
3. For every question choose **ONE** option, (A), (B), (C) or (D), and record it against the question number in the answer script provided by the invigilator.
4. Calculators are **not** allowed. Rough work may be done on separate sheets.
5. Unless a question says otherwise, use the **conventions** printed below; they apply to every section.
6. The detailed solutions are in the separate *Answer Book*; do not consult it before finishing the test.

Marking scheme (every question)

Each question has **four** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

Full marks:	+3 if ONLY the correct option is chosen;
Zero marks:	0 if none of the options is chosen (question unanswered);
Negative marks:	−1 in all other cases (wrong option, or more than one option chosen).

Conventions and notation used throughout the paper

- Linear system $Ax = b$, A a $d \times d$ matrix; $|b\rangle = b/\|b\|$, $|x\rangle = x/\|x\|$. For Hermitian A: $A = \sum_j \lambda_j |u_j\rangle\langle u_j|$, $\beta_j = \langle u_j|b\rangle$; $\kappa = |\lambda|_{\max}/|\lambda|_{\min}$ (for general matrices $\sigma_{\max}/\sigma_{\min}$); sparsity s = maximal number of non-zeros per row.
- Basis states are written with the **most significant bit first**; $\mathrm{QFT}\,|k\rangle = 2^{-n/2}\sum_y e^{2\pi i k y/2^n}|y\rangle$; $R_Y(\theta) = e^{-i\theta Y/2}$.
- **Phase estimation** with t counting qubits on $U|u\rangle = e^{2\pi i\varphi}|u\rangle$: Hadamards, controlled U^{2^j}, inverse QFT; outcome $y \approx 2^t\varphi$.
- **HHL in the notebook** (`HHL.ipynb`): n_c clock qubits, $n_b = \log_2 d$ system qubits, 1 ancilla; $U = e^{iAt}$; clock value of λ: $k = \lambda t\, 2^{n_c}/(2\pi)$, signed (values $k \geq 2^{n_c-1}$ mean $k - 2^{n_c}$); $C = 2\pi/(2^{n_c}t)$; inversion $R_Y(2\arcsin(C/\lambda_k))$ on the ancilla; success = ancilla 1 and clock 0, with $P_{\text{success}} = C^2\|A^{-1}b\|^2/\|b\|^2$.
- **Default example:** $A = H\,\mathrm{diag}(1,2,3,4)H$ with $H = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}$, $b = (1,0,0,0)$, $n_c = 4$, $t = \pi/8$, $C = 1$, 7 qubits. A *previously recorded run* means the outputs stored in the notebook (CUDA-Q 0.16.0, CPU target `qpp-cpu`).

- Numerical answers are rounded.

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SECTION A · Linear Algebra and Mathematical Foundations Q.1–20

Syllabus: linear systems, eigen-decomposition, Hermitian, unitary and positive-definite matrices, matrix functions, condition numbers, sparsity, normalisation and classical solvers.
Formats: Single-correct MCQ (Q.1–19); Matching lists (Q.3–4); Assertion–Reason (Q.7–20); Statement I / II (Q.8); Paragraph based (Q.14–15). Each question: +3 / 0 / −1.

Q.1 The solution of $\begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix} x = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ is

(A) $x = (0.5,\ 0.5)^{\mathsf{T}}$ **(B)** $x = (0.5,\ -0.5)^{\mathsf{T}}$
(C) $x = (0.25,\ -0.25)^{\mathsf{T}}$ **(D)** $x = (1,\ -1)^{\mathsf{T}}$

Q.2 The eigenvalues and the condition number $\kappa = |\lambda|_{\max}/|\lambda|_{\min}$ of $A = \begin{pmatrix} 1 & -\frac{1}{3} \\ -\frac{1}{3} & 1 \end{pmatrix}$ are

(A) $\frac{2}{3}$ and $\frac{4}{3}$; $\kappa = \frac{2}{3}$ **(B)** $\frac{8}{9}$ and 1; $\kappa = \frac{9}{8}$
(C) 1 and $-\frac{1}{3}$; $\kappa = 3$ **(D)** $\frac{2}{3}$ and $\frac{4}{3}$; $\kappa = 2$

Q.3 Match each class of square matrix (List-I) with the property of its eigenvalues (List-II).

List-I		List-II	
(P)	Hermitian ($A = A^{\dagger}$)	**(1)**	at least one eigenvalue is 0
(Q)	unitary ($U^{\dagger}U = I$)	**(2)**	all eigenvalues are real
(R)	Hermitian positive definite	**(3)**	all eigenvalues lie on the unit circle, $\lvert\lambda\rvert = 1$
(S)	singular	**(4)**	all eigenvalues are real and strictly positive

(A) P→2, Q→4, R→3, S→1 **(B)** P→2, Q→3, R→1, S→4
(C) P→2, Q→3, R→4, S→1 **(D)** P→3, Q→2, R→4, S→1

Q.4 Match each matrix (List-I) with its condition number $\kappa = |\lambda|_{\max}/|\lambda|_{\min}$ (List-II).

List-I		List-II	
(P)	diag(0.5, 2, 8)	**(1)**	4
(Q)	$\left(\begin{smallmatrix} 1 & -1/3 \\ -1/3 & 1 \end{smallmatrix}\right)$	**(2)**	1
(R)	the notebook's default A (eigenvalues $1, 2, 3, 4$)	**(3)**	16
(S)	$\left(\begin{smallmatrix} 2 & 0 \\ 0 & 2 \end{smallmatrix}\right)$	**(4)**	2

(A) P→1, Q→4, R→3, S→2 **(B)** P→3, Q→4, R→2, S→1
(C) P→3, Q→2, R→1, S→4 **(D)** P→3, Q→4, R→1, S→2

Q.5 The notebook's default matrix is $A = H\ \mathrm{diag}(1,2,3,4)\ H$ with the symmetric orthogonal matrix $H = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}$ ($H^2 = I$). For $b = (1,0,0,0)^{\mathsf{T}}$, the first component x_0 of $x = A^{-1}b$ is

(A) $\frac{5}{8}$ **(B)** $\frac{1}{4}$ **(C)** $\frac{25}{12} \approx 2.0833$ **(D)** $\frac{25}{48} \approx 0.5208$

Q.6 HHL needs a Hermitian matrix, so the notebook embeds a non-Hermitian A as $\tilde{A} = \begin{pmatrix} 0 & A \\ A^\dagger & 0 \end{pmatrix}$. For $A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$ the condition number of $\tilde{A}$ is

(A) $3 + 2\sqrt{2} \approx 5.83$ **(B)** $\sqrt{2} + 1 \approx 2.41$
(C) 1, because both eigenvalues of A are 1 **(D)** 2

Q.7 **Assertion (A):** Every invertible matrix has condition number $\kappa(A) = \|A\| \, \|A^{-1}\| \geq 1$.
Reason (R): In the spectral norm $\kappa(A) = \sigma_{\max}/\sigma_{\min}$, the ratio of the largest to the smallest singular value.

(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A
(B) Both A and R are true, but R is *not* the correct explanation of A
(C) A is true, but R is false
(D) A is false, but R is true

Q.8 **Statement I:** Every invertible matrix is Hermitian.
Statement II: Every real symmetric matrix can be written as $A = Q\Lambda Q^{\mathsf{T}}$ with an orthogonal Q and a real diagonal Λ.

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true, but Statement II is false
(D) Statement I is false, but Statement II is true

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Q.9 The sparsity s of a matrix is the largest number of non-zero entries in any row. For the notebook's default 4×4 matrix $A = \begin{pmatrix} 2.5 & -0.5 & -1 & 0 \\ -0.5 & 2.5 & 0 & -1 \\ -1 & 0 & 2.5 & -0.5 \\ 0 & -1 & -0.5 & 2.5 \end{pmatrix}$ and for a tridiagonal 1000×1000 matrix, s is

(A) 3 and 1000 **(B)** 12 and 2998 **(C)** 3 and 3 **(D)** 4 and 3

Q.10 Gaussian elimination on a dense $d \times d$ system costs about $\frac{2}{3}d^3$ floating-point operations. Doubling d from 1000 to 2000 multiplies the work by

(A) 8 **(B)** 2 **(C)** 4 **(D)** $\frac{2}{3}$

Q.11 For a sparse symmetric positive-definite matrix with condition number κ, the number of conjugate-gradient iterations needed for a fixed accuracy scales as

(A) $O(\log \kappa)$ **(B)** $O(d^3)$ **(C)** $O(\sqrt{\kappa})$ **(D)** $O(\kappa^2)$

Q.12 The vector $b = (1, 2, 2, 0)^{\mathsf{T}}$ is amplitude-encoded on two qubits as $|b\rangle = \sum_i b_i \, |i\rangle \, / \|b\|$ (index i in binary, first qubit most significant). The probability of measuring $|01\rangle$ is

(A) $\frac{1}{3}$ **(B)** $\frac{2}{9}$ **(C)** $\frac{2}{3}$ **(D)** $\frac{4}{9}$

Q.13 For the notebook's default $A = H\,\mathrm{diag}(1,2,3,4)H$ (eigenvectors = columns of H) and $b = (1,0,0,0)$, the coefficients $\beta_j = \langle u_j|b\rangle$ of $|b\rangle$ in the eigenbasis are

(A) $(1,0,0,0)$ **(B)** $(\frac{1}{2},\frac{1}{2},\frac{1}{2},\frac{1}{2})$
(C) $(\frac{1}{2},-\frac{1}{2},\frac{1}{2},-\frac{1}{2})$ **(D)** $(\frac{1}{4},\frac{1}{4},\frac{1}{4},\frac{1}{4})$

Paragraph for Questions 14 and 15

The notebook's 2×2 example is $A = \begin{pmatrix} 1 & -\frac{1}{3} \\ -\frac{1}{3} & 1 \end{pmatrix}$, $b = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$. Its eigenvectors are $u_1 = \frac{1}{\sqrt{2}}(1,1)^{\mathsf{T}}$ and $u_2 = \frac{1}{\sqrt{2}}(1,-1)^{\mathsf{T}}$.

Q.14 The solution $x = A^{-1}b$ is

(A) $(\frac{9}{8},\ \frac{3}{8})^{\mathsf{T}}$ **(B)** $(1,\ \frac{1}{3})^{\mathsf{T}}$ **(C)** $(\frac{3}{4},\ \frac{1}{4})^{\mathsf{T}}$ **(D)** $(\frac{9}{8},\ -\frac{3}{8})^{\mathsf{T}}$

Q.15 HHL prepares the normalised state $|x\rangle = x/\|x\|$. Its amplitudes, and the probability of measuring $|0\rangle$ on the system qubit, are

(A) $\frac{1}{2}(\sqrt{3},1)$ and 0.75 **(B)** $(\frac{9}{8},\frac{3}{8})$ and $\frac{9}{8}$
(C) $\frac{1}{\sqrt{10}}(3,1)$ and 0.9 **(D)** $\frac{1}{\sqrt{10}}(3,1)$ and 0.75

Q.16 For $A = \mathrm{diag}(1,2)$ and $t = \pi/2$, the matrix e^{iAt} is

(A) $\mathrm{diag}(i,\ 1)$ **(B)** $\mathrm{diag}(i,\ -1)$ **(C)** $\begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$ **(D)** $\mathrm{diag}(-i,\ -1)$

Q.17 For which matrices A is e^{iAt} unitary for every real t?

(A) exactly the sparse matrices **(B)** exactly the Hermitian matrices
(C) exactly the invertible matrices **(D)** exactly the real matrices

Q.18 For the default system (A with eigenvalues $1,2,3,4$ and $\|b\| = 1$), the norm $\|x\| = \|A^{-1}b\|$ is 0.5966. Which bound must every such norm satisfy?

(A) $\dfrac{\|b\|}{\lambda_{\max}} \le \|x\| \le \dfrac{\|b\|}{\lambda_{\min}}$, i.e. $0.25 \le 0.5966 \le 1$
(B) $\|x\| = \|b\|/\kappa = 0.25$ exactly
(C) $\|x\| \le \|b\|/\lambda_{\max} = 0.25$
(D) $\|b\|\,\lambda_{\min} \le \|x\| \le \|b\|\,\lambda_{\max}$, i.e. $1 \le \|x\| \le 4$

Q.19 For which values of a is $\begin{pmatrix} a & 1 \\ 1 & a \end{pmatrix}$ singular (so that HHL cannot be applied)?

(A) $a = \pm 2$ **(B)** $a = \pm 1$ **(C)** $a = 0$ **(D)** no real a

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Q.20 **Assertion (A):** Multiplying an invertible matrix by a non-zero constant c does not change its condition number.

Reason (R): Multiplying A by c multiplies every eigenvalue by c^2.

(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A

(B) Both A and R are true, but R is *not* the correct explanation of A

(C) A is true, but R is false

(D) A is false, but R is true

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SECTION B · Quantum Computing Foundations Q.21–40

Syllabus: amplitude encoding, measurement and post-selection, controlled gates and phase kickback, QFT and inverse QFT, quantum phase estimation, two's complement clock values, rotations, entanglement and memory.
Formats: Single-correct MCQ (Q.21–40); Matching lists (Q.23–30); Assertion–Reason (Q.35); Statement I / II (Q.29–39); Paragraph based (Q.37–38). Each question: +3 / 0 / −1.

Q.21 Phase estimation with 4 counting qubits is applied to an eigenstate of U with eigenvalue $e^{2\pi i \cdot 3/16}$. The measured counting register is

(A) $|0011\rangle$ with probability $\frac{3}{16}$
(B) $|0011\rangle$ or $|0100\rangle$, each with probability $\frac{1}{2}$
(C) $|0011\rangle$ with probability 1
(D) $|1100\rangle$ with probability 1

Q.22 In HHL, phase estimation uses $U = e^{iAt}$, so an eigenvalue λ gives the clock value $k = \lambda t\, 2^{n_c}/(2\pi)$. With $n_c = 4$ and $t = \pi/8$ (the notebook's default), the eigenvalue $\lambda = 4$ gives

(A) $k = 2$ **(B)** $k = 4\pi$
(C) $k = 8$ **(D)** $k = 4$, i.e. $|0100\rangle$

Q.23 Match each step of quantum phase estimation (List-I) with its effect (List-II).

List-I		List-II	
(P)	Hadamards on the counting register	**(1)**	converts the pattern of phases $e^{2\pi i 2^j \varphi}$ into the binary number $\approx 2^t\varphi$
(Q)	controlled U^{2^j} gates	**(2)**	returns one estimate y of $2^t\varphi$
(R)	inverse QFT on the counting register	**(3)**	creates the uniform superposition $\frac{1}{\sqrt{2^t}}\sum_j \lvert j\rangle$
(S)	measurement of the counting register	**(4)**	write the phase $e^{2\pi i 2^j \varphi}$ onto counting qubit j (phase kickback)

(A) P→3, Q→1, R→4, S→2 **(B)** P→3, Q→4, R→2, S→1
(C) P→4, Q→3, R→1, S→2 **(D)** P→3, Q→4, R→1, S→2

Q.24 A control qubit in $|+\rangle$ and a target in an eigenstate $|u\rangle$ of U ($U|u\rangle = e^{i\phi}|u\rangle$) pass through a controlled-U. The result is

(A) $\frac{1}{\sqrt{2}}(|0\rangle + e^{i\phi}|1\rangle) \otimes |u\rangle$
(B) $\frac{1}{\sqrt{2}}(|0\rangle|u\rangle + |1\rangle|u'\rangle)$ with $|u'\rangle$ orthogonal to $|u\rangle$
(C) $\frac{1}{\sqrt{2}}(e^{i\phi}|0\rangle + |1\rangle) \otimes |u\rangle$
(D) $|+\rangle \otimes e^{i\phi}|u\rangle$

Q.25 The inverse QFT on two qubits is applied to $\frac{1}{2}(|00\rangle + i|01\rangle - |10\rangle - i|11\rangle)$. The result is

(A) $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$ **(B)** $|01\rangle$
(C) $|11\rangle$ **(D)** $|10\rangle$

Q.26 The notebook builds the inverse QFT (in phase estimation) and the QFT (in the uncomputation) on $n_c = 4$ clock qubits from controlled phase gates `cr1` and `swap`s. How many `cr1` and `swap` gates does the whole HHL circuit contain?

(A) 12 and 8 **(B)** 16 and 4 **(C)** 6 and 2 **(D)** 12 and 4

Q.27 An ancilla in $|0\rangle$ is rotated by $R_Y(\theta) = e^{-i\theta Y/2}$ with $\theta = 2\arcsin(\frac{1}{2})$. The probability of then measuring $|1\rangle$ is

(A) $\frac{3}{4}$ **(B)** $\frac{1}{4}$ **(C)** $\sin^2(\frac{1}{2}) \approx 0.23$ **(D)** $\frac{1}{2}$

Q.28 To apply a rotation only when the 4-qubit clock register is in $|k\rangle$, the notebook surrounds a fully controlled `cry` with X gates on the clock qubits whose bit of k is 0 (clock[0] = most significant bit). For $k = 5$, the X gates act on

(A) clock[2] and clock[3] **(B)** clock[0] and clock[2]
(C) clock[0] only **(D)** clock[1] and clock[3]

Q.29 **Statement I:** A controlled-U^{2^j} can be implemented as 2^j consecutive controlled-U gates.
Statement II: For $U = e^{iAt}$, $U^{2^j} = e^{iAt\,2^j}$.

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true, but Statement II is false
(D) Statement I is false, but Statement II is true

Q.30 HHL reads the 4-qubit clock as a signed (two's complement) number: values $k \geq 8$ mean $k - 16$. Match each clock value (List-I) with the signed value (List-II).

List-I	List-II
(P) $k = 3$	**(1)** -1
(Q) $k = 8$	**(2)** -4
(R) $k = 12$	**(3)** 3
(S) $k = 15$	**(4)** -8

(A) P→3, Q→2, R→4, S→1 **(B)** P→3, Q→4, R→1, S→2
(C) P→3, Q→4, R→2, S→1 **(D)** P→1, Q→4, R→2, S→3

Q.31 A state $|\Psi\rangle = \sqrt{1-p}\,|\psi_0\rangle\,|0\rangle + \sqrt{p}\,|\psi_1\rangle\,|1\rangle$ (normalised $|\psi_0\rangle$, $|\psi_1\rangle$) is measured on the last qubit and the result 1 is kept (post-selection). Which statement is correct?

(A) The outcome 1 occurs with probability p and leaves the rest in $|\psi_1\rangle$
(B) The outcome 1 occurs with probability $\sqrt{p}$
(C) Post-selection succeeds with probability 1 if $p > 0$
(D) The remaining state is $\sqrt{p}\,|\psi_1\rangle$, not normalised

Q.32 After phase estimation in HHL the system and clock registers are in $\sum_j \beta_j |u_j\rangle |k_j\rangle$ (distinct k_j). When is this a product state of the two registers?

(A) always

(B) never

(C) when $|b\rangle$ is an eigenvector of A (only one $\beta_j \neq 0$)

(D) when all β_j are equal

Q.33 A state vector of n qubits needs 16×2^n bytes (fp64). The notebook's default HHL circuit has 7 qubits, yet it prints "Estimated memory: 4.10 KB". Why?

(A) The density matrix of 7 qubits needs exactly 4.10 KB

(B) Each qubit needs $16 \times 7 = 112$ B and there are 36 of them

(C) The state needs 2048 B and the notebook reserves one extra copy (for reading the full state), giving 4096 B

(D) Custom gates add 2048 B of matrix storage per qubit

Q.34 Preparing an arbitrary normalised n-qubit amplitude vector $|b\rangle$ from $|0 \ldots 0\rangle$ with one- and two-qubit gates requires in general

(A) $O(n)$ gates **(B)** $O(\log n)$ gates

(C) a single gate per qubit **(D)** $O(2^n)$ gates

Q.35 **Assertion (A):** In HHL, measuring the clock register immediately after phase estimation would destroy the superposition over eigencomponents that $|x\rangle$ needs.

Reason (R): The inverse QFT is a unitary operation.

(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A

(B) Both A and R are true, but R is *not* the correct explanation of A

(C) A is true, but R is false

(D) A is false, but R is true

Q.36 A swap test between states $|\psi\rangle$ and $|\phi\rangle$ gives the ancilla outcome 0 with probability $P(0) = \frac{1}{2}(1 + |\langle\psi|\phi\rangle|^2)$. If $P(0) = 0.9$ is measured, the overlap $|\langle\psi|\phi\rangle|^2$ is

(A) 0.8 **(B)** 0.95 **(C)** 0.9 **(D)** 0.4

Paragraph for Questions 37 and 38

Quantum phase estimation with $t = 2$ counting qubits is applied to an eigenstate $|u\rangle$ of a unitary U, $U|u\rangle = e^{2\pi i\varphi}|u\rangle$. The counting register ends in $\sum_y \alpha_y |y\rangle$ with $\alpha_y = \frac{1}{4}\sum_{j=0}^{3} e^{2\pi i j(\varphi - y/4)}$, and the outcome y estimates $\varphi \approx y/4$.

Q.37 For $\varphi = \frac{1}{4}$, the measured counting register is

(A) $|10\rangle$ with probability 1
(B) $|01\rangle$ with probability 1
(C) $|01\rangle$ with probability $\frac{1}{4}$
(D) $|00\rangle$ with probability 1

Q.38 For $\varphi = \frac{1}{3}$, the most likely outcome and its probability are

(A) all four outcomes, 0.25 each
(B) $|01\rangle$, exactly 1
(C) $|10\rangle$, about 0.70
(D) $|01\rangle$, about 0.70

Q.39 **Statement I:** The inverse QFT is the adjoint of the QFT, so applying it after the QFT returns the original state.

Statement II: The QFT on n qubits requires a number of gates exponential in n.

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true, but Statement II is false
(D) Statement I is false, but Statement II is true

Q.40 With $n_c = 4$ and $t = \pi/8$, the matrix A is replaced by $A + I$. How do the ideal clock values change?

(A) each clock value increases by 1
(B) each clock value increases by $\pi/8$
(C) they are unchanged, because I only adds a global phase
(D) each clock value doubles

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SECTION C · HHL Algorithm Fundamentals Q.41–60

Syllabus: registers, state preparation, phase estimation with e^{iAt}, eigenvalue inversion and the constant C, controlled rotations, uncomputation, post-selection, norm recovery and the notebook's circuit and code.
Formats: Single-correct MCQ (Q.41–57); Matching lists (Q.46–58); Assertion–Reason (Q.49); Statement I / II (Q.52); Paragraph based (Q.59–60). Each question: +3 / 0 / −1.

Q.41 After phase estimation and the controlled rotation (before uncomputation), the ideal HHL state is (system ⊗ clock ⊗ ancilla)

(A) $\sum_j \beta_j |u_j\rangle |0\rangle \left(\sqrt{1-\frac{C^2}{\lambda_j^2}}\,|0\rangle + \frac{C}{\lambda_j}|1\rangle\right)$ **(B)** $\sum_j \frac{\beta_j}{\lambda_j} |u_j\rangle |0\rangle |1\rangle$

(C) $\sum_j \beta_j |u_j\rangle |k_j\rangle \left(\sqrt{1-C^2\lambda_j^2}\,|0\rangle + C\lambda_j|1\rangle\right)$ **(D)** $\sum_j \beta_j |u_j\rangle |k_j\rangle \left(\sqrt{1-\frac{C^2}{\lambda_j^2}}\,|0\rangle + \frac{C}{\lambda_j}|1\rangle\right)$

Q.42 Using the notebook's layout (clock + system + one ancilla), an 8×8 Hermitian system with $n_c = 5$ clock qubits needs how many qubits?

(A) 14 **(B)** 9 **(C)** 13 **(D)** 8

Q.43 The notebook sets $C = 2\pi/(2^{n_c}t)$, the smallest non-zero eigenvalue magnitude the clock can represent. Why must $C \le |\lambda_j|$ for every eigenvalue?

(A) The rotation amplitude C/λ_j must satisfy $|C/\lambda_j| \le 1$ so that $\arcsin(C/\lambda_j)$ exists
(B) Otherwise phase estimation needs more counting qubits than system qubits
(C) C must be the largest eigenvalue so that the success probability is 1
(D) Otherwise e^{iAt} would not be unitary

Q.44 With $C = 1$, the eigenvalue-inversion rotation for the clock value of $\lambda = 2$ is $R_Y(\theta)$ with

(A) $\theta = \frac{\pi}{2}$ **(B)** $\theta = \arcsin\frac{1}{2} = \frac{\pi}{6}$ **(C)** $\theta = 2\arcsin 2$ **(D)** $\theta = 2\arcsin\frac{1}{2} = \frac{\pi}{3}$

Q.45 For the default system ($\beta_j = \frac{1}{2}$ for the eigenvalues $1, 2, 3, 4$; $C = 1$), the ideal success probability $P_{\text{success}} = \sum_j |\beta_j|^2 C^2/\lambda_j^2$ is

(A) 1.4236 **(B)** 0.25 **(C)** 0.3559 **(D)** 0.5208

Q.46 Match each stage of the notebook's HHL circuit (List-I) with what it does to the registers (List-II).

List-I		List-II	
(P)	amplitude initialisation	**(1)**	$\lvert u_j\rangle \lvert k_j\rangle \lvert 0\rangle \to \lvert u_j\rangle \lvert k_j\rangle \left(\sqrt{1-C^2/\lambda_j^2}\,\lvert 0\rangle + \frac{C}{\lambda_j}\lvert 1\rangle\right)$
(Q)	phase estimation	**(2)**	returns the clock register to $\lvert 0\ldots0\rangle$ so that it is no longer entangled with the system
(R)	eigenvalue inversion	**(3)**	prepares $\lvert 0\ldots0\rangle_{\text{clock}} \lvert b\rangle \lvert 0\rangle_{\text{anc}}$
(S)	uncomputation	**(4)**	$\lvert b\rangle \lvert 0\rangle \to \sum_j \beta_j \lvert u_j\rangle \lvert k_j\rangle$

(A) P→3, Q→4, R→1, S→2 **(B)** P→3, Q→4, R→2, S→1
(C) P→4, Q→3, R→1, S→2 **(D)** P→3, Q→1, R→4, S→2

Q.47 With `EVOLUTION_TIME = "auto"` the notebook chooses $t = 2\pi/(2^{n_c}|\lambda|_{\min})$ whenever $\kappa \le 2^{n_c-1} - 1$. For $n_c = 4$ and eigenvalues $0.5, 1, 2, 3$, it picks

(A) $t = \pi/8$ and $C = 1$ **(B)** $t = 2\pi/3$ and $C = 3$

(C) $t = \pi/4$ and $C = 1$ **(D)** $t = \pi/4$ and $C = 0.5$

Q.48 Why does HHL uncompute phase estimation after the eigenvalue inversion instead of simply measuring the ancilla?

(A) Uncomputation doubles the success probability

(B) Uncomputation is needed to make e^{iAt} unitary

(C) It converts the amplitudes C/λ_j into probabilities

(D) Without it the clock stays entangled with the system, so the post-selected system register would be a mixture instead of $|x\rangle$ (unless the clock is also post-selected on a fixed value)

Q.49 **Assertion (A):** From the simulated state vector, the notebook recovers the full solution x including its norm.

Reason (R): In the post-selected branch (clock = 0, ancilla = 1) the system amplitudes are $\frac{C}{\|b\|}\,x$, so $x = \frac{\|b\|}{C} \times$ (amplitudes).

(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A

(B) Both A and R are true, but R is *not* the correct explanation of A

(C) A is true, but R is false

(D) A is false, but R is true

Q.50 The notebook's eigenvalue inversion applies one multi-controlled rotation for every non-zero clock value. With n_c clock qubits this is

(A) n_c rotations **(B)** $2^{n_c} - 1$ rotations (15 for $n_c = 4$)

(C) one rotation per eigenvalue of A **(D)** 2^{n_c} rotations, including $k = 0$

Q.51 With $n_c = 4$ and $t = \pi/8$, the notebook assigns to the clock value $k = 12$ the eigenvalue

(A) 12 **(B)** -4 **(C)** 4 **(D)** -12

Q.52 **Statement I:** If some $\lambda_j t\, 2^{n_c}/(2\pi)$ is not an integer, phase estimation spreads that component over several clock values and the fidelity of $|x\rangle$ drops below 1.

Statement II: The notebook's `"auto"` evolution time guarantees exact clock values for every matrix.

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true, but Statement II is false

(D) Statement I is false, but Statement II is true

Q.53 For the notebook's 2×2 example ($\lambda = \frac{2}{3}, \frac{4}{3}$) with $n_c = 4$ and `"auto"` time, the values of t, C and the clock values are

(A) $t = \frac{3\pi}{16}$, $C = \frac{4}{3}$, clock values 1 and 2 **(B)** $t = \frac{3\pi}{8}$, $C = \frac{2}{3}$, clock values 2 and 4

(C) $t = \frac{\pi}{8}$, $C = 1$, clock values $\frac{2}{3}$ and $\frac{4}{3}$ **(D)** $t = \frac{3\pi}{16}$, $C = \frac{2}{3}$, clock values 1 and 2

Q.54 In the sampling read-out the notebook keeps a shot only if

(A) the clock bits encode an eigenvalue

(B) the ancilla bit is 0

(C) all clock bits are 0 and the ancilla bit is 1

(D) the system bits are all 0

Q.55 The ideal success probability is $p = 0.3559$ and the notebook uses 20 000 shots. The standard error of the measured fraction of accepted shots is about

(A) 0.0034 **(B)** 0.0001 **(C)** 0.0119 **(D)** 0.048

Q.56 On hardware, the post-selected system bitstrings of HHL directly estimate

(A) $\|x\|$ only

(B) $|x_i|^2/\|x\|^2$ only — signs/phases and the norm need further measurements

(C) $x_i/\|x\|$ including its sign

(D) every component x_i including its sign

Q.57 The recorded circuit statistics of the default HHL circuit are `h: 16, ccustom_unitary_constant: 8, swap: 4, cr1: 12, x: 56, ccccry: 15`. How many multi-qubit gates is that?

(A) 111 **(B)** 39 **(C)** 23 **(D)** 95

Q.58 Match each HHL ingredient (List-I) with the code that implements it in `HHL.ipynb` (List-II).

List-I		List-II	
(P)	preparation of $\|0\rangle\,\|b\rangle\,\|0\rangle$	**(1)**	`kernel.cry(theta, clock, ancilla)` between X gates
(Q)	controlled powers U^{2^j}	**(2)**	`postselected_amplitudes(state_vector)`
(R)	eigenvalue inversion	**(3)**	`kernel.qalloc(bit_reverse_order(initial, n_tot))`
(S)	extraction of the post-selected amplitudes	**(4)**	`cudaq.register_operation(name, matrix_power(U, 2**j))`

(A) P→4, Q→3, R→1, S→2 **(B)** P→3, Q→1, R→4, S→2

(C) P→3, Q→4, R→1, S→2 **(D)** P→3, Q→4, R→2, S→1

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Paragraph for Questions 59 and 60

HHL in the notebook's convention is applied to $A = \mathrm{diag}(1, 2)$ and $b = \frac{1}{\sqrt{2}}(1, 1)$ with $n_c = 3$ clock qubits and $t = \pi/4$, so $U = e^{iAt}$, the clock value of an eigenvalue λ is $k = \lambda t\, 2^{n_c}/(2\pi) = \lambda$, and $C = 2\pi/(2^{n_c} t) = 1$. The eigenvalue inversion applies $R_Y(2 \arcsin(C/\lambda_k))$ to the ancilla for each clock value; the system register has one qubit.

Q.59 After phase estimation the state of system $\otimes$ clock (ancilla still $|0\rangle$) is

(A) $\frac{1}{\sqrt{2}}(\,|0\rangle\,|001\rangle + |1\rangle\,|011\rangle\,)$

(B) $\frac{1}{\sqrt{2}}(\,|0\rangle\,|100\rangle + |1\rangle\,|010\rangle\,)$

(C) $\frac{1}{\sqrt{2}}(\,|0\rangle\,|001\rangle + |1\rangle\,|010\rangle\,)$

(D) $\frac{1}{\sqrt{2}}(\,|0\rangle + |1\rangle\,)\,|001\rangle$

Q.60 The success probability and the normalised output state $|x\rangle$ are

(A) 0.625 and $\frac{1}{\sqrt{5}}(2, 1)$

(B) 0.625 and $\frac{1}{\sqrt{2}}(1, 1)$

(C) 0.75 and $\frac{1}{\sqrt{5}}(2, 1)$

(D) 1 and $\frac{1}{\sqrt{5}}(1, 2)$

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SECTION D · Advanced HHL and Implementation Q.61–80

Syllabus: complexity, condition number and success probability, amplitude amplification, read-out cost, clock precision and aliasing, embedding and padding, Hamiltonian simulation, newer algorithms, backends, recorded results and limitations.
Formats: Single-correct MCQ (Q.61–78); Matching lists (Q.64–72); Assertion–Reason (Q.69); Statement I / II (Q.74); Paragraph based (Q.79–80). Each question: +3 / 0 / −1.

Q.61 The original HHL analysis (sparse $d \times d$ matrix with sparsity s, condition number κ, precision ε) gives a running time of

(A) $O(\log d)$, independent of κ and ε
(B) $O(d\, s\sqrt{\kappa}\log(1/\varepsilon))$
(C) $O(d^3)$
(D) $\tilde{O}(\log(d)\, s^2\kappa^2/\varepsilon)$

Q.62 HHL succeeds (ancilla = 1) with probability $p = 0.01$. On average, how many repetitions are needed with plain repetition, and with amplitude amplification?

(A) about 100 and about 10
(B) about 100 and about 1
(C) about 1 and about 10
(D) about 10 and about 100

Q.63 To estimate all $d = 1024$ probabilities $|x_i|^2$ of $|x\rangle$ to absolute precision $\varepsilon = 0.01$ by sampling, roughly how many post-selected shots are needed?

(A) about 10^{14} **(B)** about 10 **(C)** about 10^7 **(D)** about 10^3

Q.64 For exact clock values the notebook needs $\kappa \le 2^{n_c-1} - 1$. Match each condition number (List-I) with the smallest suitable number of clock qubits n_c (List-II).

List-I	List-II
(P) $\kappa = 3$	**(1)** 6
(Q) $\kappa = 7$	**(2)** 3
(R) $\kappa = 20$	**(3)** 7
(S) $\kappa = 40$	**(4)** 4

(A) P→2, Q→4, R→3, S→1
(B) P→2, Q→1, R→4, S→3
(C) P→4, Q→2, R→1, S→3
(D) P→2, Q→4, R→1, S→3

Q.65 With `NUM_CLOCK_QUBITS = 8`, how many multi-controlled eigenvalue-inversion rotations does the notebook's circuit contain, and how many qubits does the default 4×4 system then use?

(A) 255 rotations, 11 qubits
(B) 255 rotations, 14 qubits
(C) 256 rotations, 12 qubits
(D) 8 rotations, 11 qubits

Q.66 With `MATRIX_A = [[1, 2], [0, 1]]` (not Hermitian) and $n_c = 4$, the notebook's circuit uses how many qubits, and where is x in the solution of the embedded system?

(A) 7 qubits; x is the first half of the solution
(B) 7 qubits; x is the second half of the solution
(C) 8 qubits; x is the second half
(D) 6 qubits; x is the whole solution

Q.67 A 3×3 Hermitian matrix with eigenvalues $5, 10, 20$ is padded by the notebook to 4×4 with an identity block. What happens to κ?

(A) it grows from 4 to 20, because the padding adds the eigenvalue 1

(B) it stays 4

(C) it becomes 5

(D) it becomes 1

Q.68 "Dequantised" algorithms (Tang, 2018 and later) showed that for some quantum linear-algebra tasks

(A) HHL can run without phase estimation

(B) the condition number no longer affects quantum algorithms

(C) classical computers can prepare $|x\rangle$ on a quantum computer

(D) classical randomized algorithms with comparable sampling access achieve similar poly-logarithmic dependence on d for low-rank matrices

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Q.69 **Assertion (A):** HHL gives an exponential speed-up over classical algorithms for every linear system $Ax = b$.

Reason (R): For s-sparse, well-conditioned A with efficiently preparable $|b\rangle$, HHL prepares $|x\rangle$ in time polynomial in $\log d$, s, κ and $1/\varepsilon$.

(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A

(B) Both A and R are true, but R is *not* the correct explanation of A

(C) A is true, but R is false

(D) A is false, but R is true

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Q.70 Take $C = \lambda_{\min}$ and a matrix with $\kappa = 20$. If $|b\rangle$ is the eigenvector of $\lambda_{\max}$, the success probability is

(A) 0.5 **(B)** $1/\kappa^2 = 0.0025$ **(C)** 1 **(D)** $1/\kappa = 0.05$

Q.71 With $n_c = 4$ and $t = \pi/8$ (clock value $k = \lambda$), how is an eigenvalue $\lambda = 9$ treated?

(A) Phase estimation fails with an error

(B) It is read as 7, the largest positive value

(C) It aliases: clock value 9 is read as the signed value -7, so the wrong (negative) $1/\lambda$ is applied

(D) It is read correctly as 9

Q.72 Match each quantum linear-system algorithm (List-I) with its key idea (List-II).

List-I		List-II	
(P)	HHL (2009)	**(1)**	implement a polynomial approximation of $1/x$ on the singular values of a block-encoded A
(Q)	Ambainis (2012), variable-time amplitude amplification	**(2)**	phase estimation + controlled rotation C/λ + post-selection; cost κ^2/ε
(R)	Childs–Kothari–Somma (2017)	**(3)**	amplify different eigenvalue branches at different times: κ dependence close to linear ($\kappa \log^3 \kappa$)
(S)	QSVT-based solvers	**(4)**	linear combination of unitaries (Fourier/Chebyshev series of $1/x$): $\mathrm{poly}\log(1/\varepsilon)$ precision dependence

(A) P→2, Q→4, R→3, S→1 **(B)** P→3, Q→2, R→4, S→1
(C) P→2, Q→3, R→4, S→1 **(D)** P→2, Q→3, R→1, S→4

Q.73 With `SIMULATION_METHOD = "density_matrix"` the notebook multiplies its recovered x by a phase factor before comparing with NumPy. Why?

(A) A density matrix $|\psi\rangle\langle\psi|$ carries no global phase, so the vector extracted from it has an arbitrary phase that must be aligned

(B) The density-matrix simulator uses fp32 precision

(C) The density-matrix simulator adds noise that rotates the phase

(D) Post-selection on a density matrix flips the sign of every component

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Q.74 **Statement I:** `HHL.ipynb` refuses hardware backends, because exact matrix exponentials as custom gates and amplitude initialisation cannot run on hardware targets.

Statement II: The controlled powers U^{2^j} are registered as custom unitary matrices with `cudaq.register_operation`.

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true, but Statement II is false

(D) Statement I is false, but Statement II is true

Q.75 Run with only `NUM_CLOCK_QUBITS = 3` on the default system ($\kappa = 4 > 3$ levels), the notebook rescales t so that $\lambda_{\max} = 4$ maps to clock value 3. The ideal clock values of $\lambda = 1, 2, 3, 4$ and the effect are

(A) $1, 2, 3, 4$ — still exact

(B) $0.75, 1.5, 2.25, 3$ — non-integer values are rounded by phase estimation, so the fidelity drops below 1

(C) $3, 6, 9, 12$ — all eigenvalues alias

(D) $0.75, 1.5, 2.25, 3$ — the fidelity stays exactly 1

Q.76 With `NUM_CLOCK_QUBITS = 8` the default circuit has 11 qubits. The fp64 state vector needs ... bytes, and the notebook's memory line (state + one host copy) shows about ...

(A) 32 768; 32.8 kB **(B)** 32 768; 65.5 kB **(C)** 4 194 304; 8.4 MB **(D)** 2 048; 4.1 kB

Q.77 Which statement correctly describes how `HHL.ipynb` implements $U = e^{iAt}$?

(A) It uses qubitisation with a block encoding of A

(B) It uses first-order Trotterisation of A's Pauli decomposition

(C) It computes e^{iAt} exactly with `scipy.linalg.expm` and applies each power U^{2^j} as a custom controlled gate; this is exact on simulators but is not a scalable Hamiltonian simulation

(D) It approximates e^{iAt} by $I + iAt$

Q.78 Which use of HHL can, in principle, keep an exponential advantage?

(A) replacing Gaussian elimination for 4×4 systems

(B) printing every component of x for a dense $10^6 \times 10^6$ matrix

(C) solving a dense, ill-conditioned system with $\kappa \sim d$

(D) estimating a quantity such as $x^\dagger M x$ for a sparse, well-conditioned A with efficiently preparable $|b\rangle$

Paragraph for Questions 79 and 80

Previously recorded default run of ***`HHL.ipynb`*** (CUDA-Q 0.16.0, target `qpp-cpu`): 4×4 system, $\kappa = 4$, 7 qubits, $\|b\| = 1$, $C = 1$. Success probability: 0.355903 from the state, 0.3543 from 20 000 shots (7086 accepted). Exact $|x_i|^2/\|x\|^2 = (0.7622, 0.0598, 0.1476, 0.0305)$; from the accepted shots $(0.7691, 0.0587, 0.1420, 0.0302)$. Fidelity of the state read-out 1.0000000000, relative error 1.655×10^{-15}.

Q.79 The sampled $|x_0|^2/\|x\|^2 = 0.7691$ differs from the exact 0.7622 by 0.0069. With 7086 accepted shots, its binomial standard error is about ..., so the deviation is ...

(A) 0.0051; about 14 standard errors — a bug

(B) 0.0051; about 1.4 standard errors — consistent with shot noise

(C) 0.0034; about 2 standard errors

(D) 0.0119; about 0.6 standard errors

Q.80 Using $P_{\text{success}} = C^2\|x\|^2/\|b\|^2$, the norm $\|x\|$ estimated from the shot-based success fraction is about

(A) 1.68 **(B)** 0.354 **(C)** 0.595 **(D)** 0.125